End-to-End Data Management Pipeline for a Recommendation System

Data Management for Machine Learning (DM4ML) - Assignment 2 · BITS Pilani WILP, M.Tech AI/ML

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GitHub repository: https://github.com/vssr32000/recommendation-pipeline

1. Problem and approach

The task is to recommend movies to users of an online movie platform from their past ratings and item metadata, and, more importantly, to build the **data management pipeline** around the model so that it is validated, reproducible and automated. I used the public **MovieLens-1M** dataset (6,040 users with demographics, 3,883 movies with genres, 1,000,209 timestamped 1-5 star ratings). The pipeline has seven stages, each a separate Python module with a `run(cfg)` entry point, driven by one YAML config and orchestrated as a Prefect flow (Figure 1).

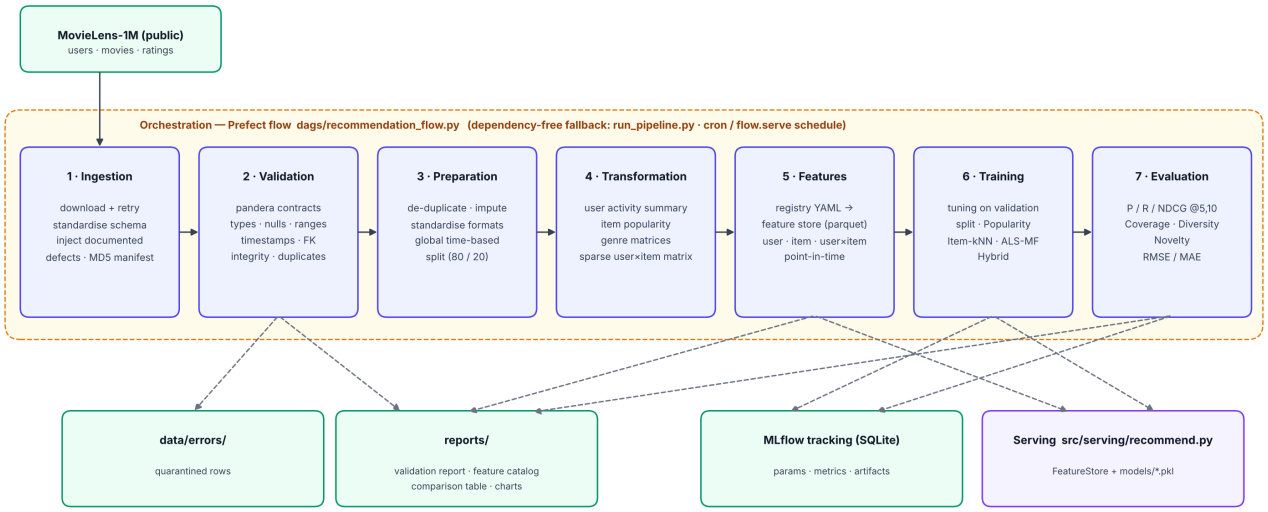


Figure 1 - Pipeline architecture. Solid arrows: task dependencies; dashed arrows: outputs.

2. Design decisions and tools

Stage	Tool(s)	Key design decision
Ingestion	urllib, pandas	Download with retries and a local cache; standardise all sources into one users / items / interactions contract; MD5 + row-count manifest for lineage. Because MovieLens is already clean, 880 seeded, logged defects are injected so validation can be tested against a known answer.
Validation	pandera	Declarative schemas per table. Every rule has a severity: *error* rows (wrong type, null key, invalid rating/timestamp, orphan FK) are quarantined with reasons; *warning* rows (duplicates, missing optional attributes) pass and are repaired downstream, so no user is dropped and foreign keys stay valid. A data-quality gate fails the run above 1% rejected rows.
Preparation	pandas, parquet	Deterministic cleaning (latest re-rating wins, mode/'U'/'Unknown' imputation, UTC datetimes, parsed release year). One **global time cut-off** (80th percentile) so no future rating can leak into training, which per-user or random splits allow.
Transformation	pandas, scipy.sparse	Derived datasets come from the train split only: user activity summary, item popularity with a Bayesian-smoothed rating, genre matrices, and a CSR user-item matrix.
Features	Custom feature store (YAML registry + parquet)	One registry defines each feature; the build fails if a declared feature is missing; the catalog is generated from the registry. The same `FeatureStore` API serves training, evaluation and the serving CLI.
Training	NumPy/SciPy, MLflow	Four models behind one interface. Hyper-parameters are tuned on the last 10% of the *train* period (never the test set). MLflow logs every trial as a nested run plus params, lineage tags (cut-off, feature MD5s), loss curve and model artifacts.
Evaluation	NumPy, matplotlib	Ranking metrics at K=5 and 10 on warm test users, plus catalogue coverage, genre intra-list diversity, novelty and RMSE/MAE; metrics are written back to each MLflow run.

Stage	Tool(s)	Key design decision
Orchestration	Prefect 3 (+ plain-Python runner)	The DAG is declared once and used by both orchestrators; tasks have explicit dependencies, ingestion retries, and a cron schedule via `flow.serve`. The fallback runner skips downstream stages on failure.

3. Data validation and preparation results

Of 1,000,509 raw interactions, **520 were quarantined** (pass rate 99.948%) and 450 passed with warnings. All 6 of 6 injected error-type defect classes were detected in full. Users (43 warnings) and items (17) needed repair only, with no rejections.

Rule (interactions)	Severity	Rows	Rule (interactions)	Severity	Rows
duplicate_interaction	warning	304	missing_key	error	150
duplicate_row	warning	150	orphan_item	error	50
invalid_rating	error	120	orphan_user	error	50
invalid_timestamp	error	90	wrong_type	error	60

After cleaning, the time-based split at **2000-12-02** gives 799,725 training ratings (5,392 users, 3,662 items) and 103,783 warm test ratings from 1,135 users. 96,064 test ratings belong to users who joined after the cut-off. A personalised model cannot score these cold-start users, so they are reported and left out of the comparison; in serving they fall back to the popularity model.

4. Feature management

The registry defines **28 features** (14 user, 11 item and 3 on-demand user×item features; two of them are 19-dimensional genre vectors). All are computed *as of* the split cut-off (point-in-time correct), stored as parquet with MD5 checksums, and the checksums are logged to every MLflow training run. Examples: `user_avg_rating`, `user_activity_count`, `user_preferred_genre`, `user_days_since_last_interaction`, `item_popularity`, `item_bayes_rating`, `ui_genre_affinity`. The popularity and hybrid models read their signals from the store, and so does `src/serving/recommend.py`. The full catalog with types, sources and live statistics is in `reports/feature_catalog.md`.

5. Models and evaluation results

Popularity (baseline) recommends the most-rated unseen items. **Item-kNN** is item-based collaborative filtering with shrunk cosine similarity. **ALS-MF** is biased explicit matrix factorisation (48 factors, ALS-WR) that predicts ratings. **Hybrid** blends min-max-normalised item-kNN and ALS scores with the user×item genre affinity and item popularity features. Tuned values: item-kNN 300 neighbours with shrinkage 100; ALS lambda = 0.08; hybrid weights 0.45 / 0.25 / 0.05 / 0.25. Relevant = test rating of 4 or more; items already seen are excluded.

Model	P@10	R@10	NDCG@10	Hit@10	Cov@10	Div@10	RMSE	MAE
Popularity (baseline)	0.1986	0.0479	0.2129	0.6284	0.0374	0.7601	0.9595	0.7597
Item-kNN CF	0.1912	0.0548	0.2124	0.6221	0.1436	0.7370	-	-
ALS Matrix Factorisation	0.0954	0.0256	0.1006	0.4434	0.1671	0.7904	0.8941	0.6956
Hybrid (CF + MF + features)	0.2265	0.0646	0.2489	0.6732	0.1092	0.7443	-	-

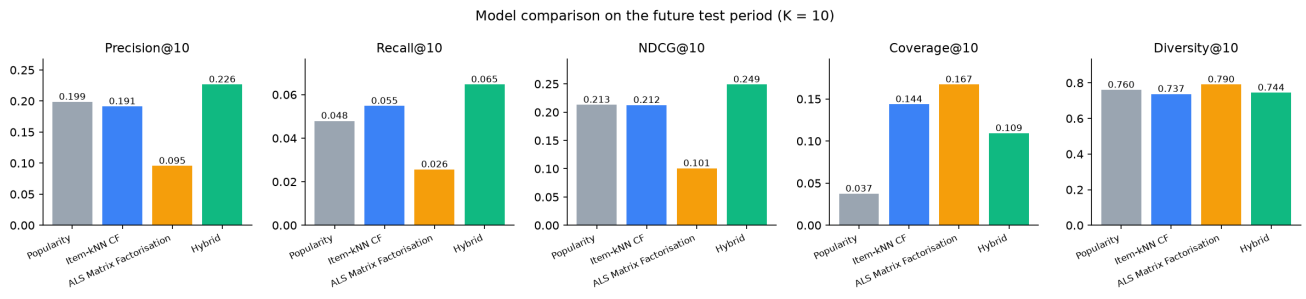


Figure 2 - Model comparison on the future test period (K = 10).

Interpretation. The **hybrid model is best on every accuracy metric**: NDCG@10 0.2489 vs 0.2129 (+16.9%), Recall@10 +35%, and about 2.9x the catalogue coverage. Popularity is a strong baseline under a temporal split, because the test period is dominated by mainstream titles. Item-kNN matches it on NDCG while covering about 4x

more items. ALS gives the best rating prediction (RMSE 0.894 vs 0.960 for the item-mean baseline), but ranking by predicted rating pushes long-tail items with high predicted scores (novelty 4.4), so its top-K accuracy is low. That is the usual gap between rating prediction and top-K ranking, and it is why ALS contributes to the hybrid as a signal and is not used on its own. Results are identical across repeated runs (fixed seed and deterministic splits): the MLflow runs table (A10) shows the same metrics for every execution.

6. Limitations and future work

• **Injected defects.** Data-quality issues are synthetic (seeded and logged), so the validation layer is proven against known defects, not against real production drift.

• **Cold start.** About half of the raw test ratings come from users who joined after the cut-off. They are served by popularity only. Demographic or content-based onboarding models could cover them.

• **Offline evaluation only.** A single temporal split on a 2000-2003 dataset. A rolling-origin backtest and online A/B metrics (CTR, watch time) would give a better estimate of real impact.

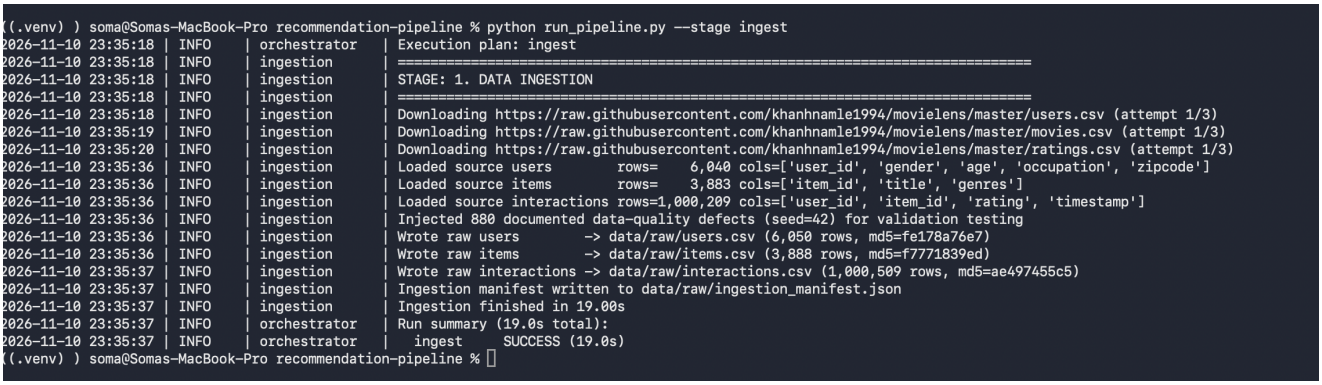
• **Scale.** The dense item-item similarity matrix and the pure-NumPy ALS loop suit 1M ratings. Larger catalogues need approximate nearest neighbours, the `implicit` library or Spark ALS.

• **Feature store.** The file-based store has no online, low-latency serving layer or TTLs. Feast with a Redis/Postgres online store is the natural next step. Hybrid tuning uses features built on the full train period, a mild optimistic bias limited to the choice of weights.

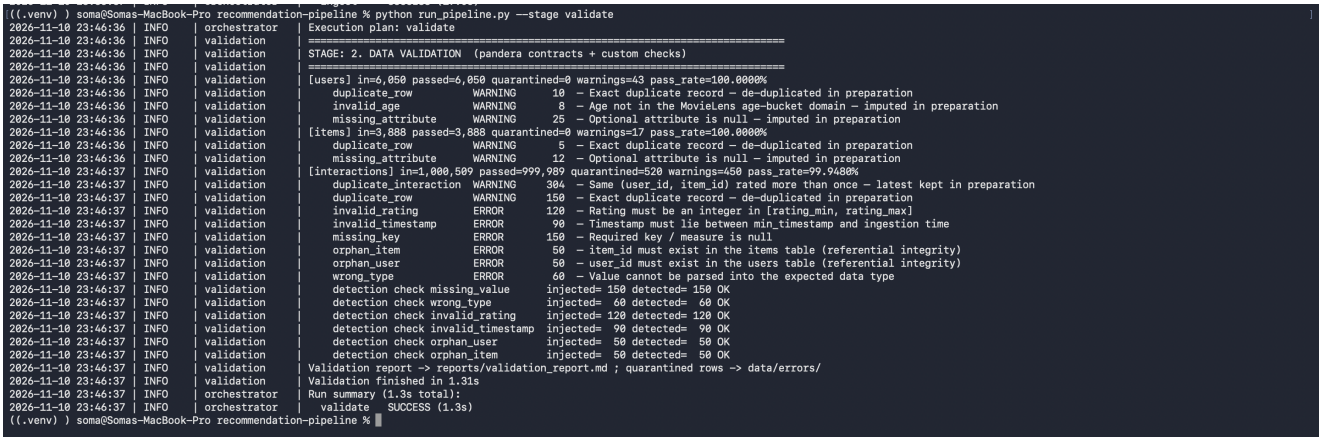
• **Model registry.** Models are tracked as MLflow artifacts, but they are not promoted through MLflow Model Registry stages, and the pipeline has no CI/CD.

Appendix A - Execution screenshots of every pipeline stage

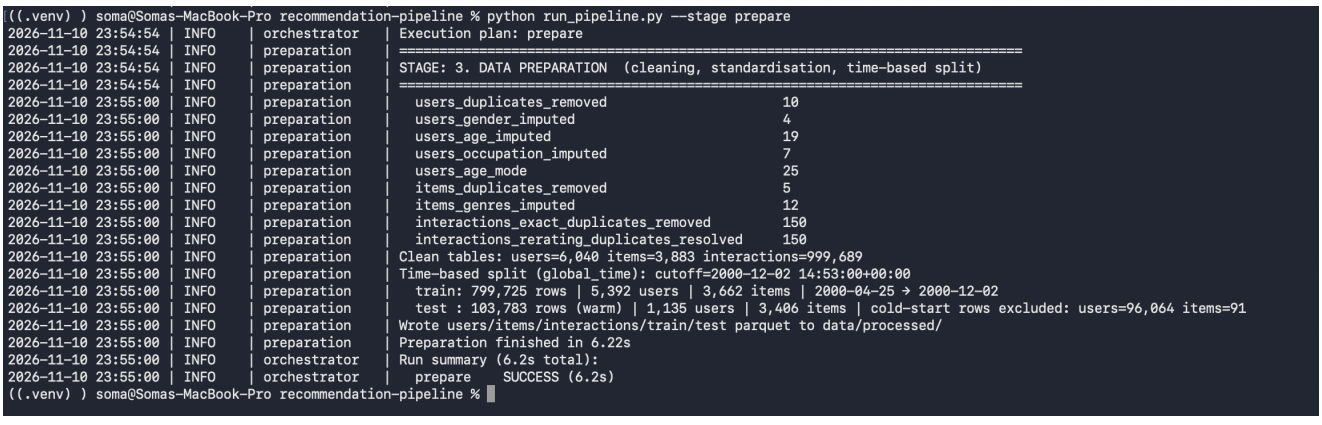
Each stage was executed individually with `python run_pipeline.py --stage <name>`, then the full DAG was executed as a Prefect flow against a local Prefect server. Terminal captures show the unedited program output (A8 shows only the Prefect orchestration lines; the complete flow output is in `reports/prefect_flow_run.log`). UI captures are from the Prefect and MLflow web UIs.



A1 - Stage 1, Data ingestion: download/cache, schema standardisation, defect injection, manifest with MD5.



A2 - Stage 2, Data validation: pandera rule counts by severity, quarantine, detection check against injected defects.



A3 - Stage 3, Data preparation: de-duplication and imputation counts, time-based train/test split.

```
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline % python run_pipeline.py --stage transform
2026-11-10 23:57:27 | INFO     | orchestrator   | Execution plan: transform
2026-11-10 23:57:27 | INFO     | transformation | ============================================================================
2026-11-10 23:57:27 | INFO     | transformation | STAGE: 4. DATA TRANSFORMATION  (derived datasets from the train split)
2026-11-10 23:57:27 | INFO     | transformation | ============================================================================
2026-11-10 23:57:28 | INFO     | transformation | user_activity_summary : 5392 rows x 10 cols
2026-11-10 23:57:28 | INFO     | transformation | item_popularity       : 3662 rows x 10 cols  (top item: American Beauty (1999), 2901 ratings)
2026-11-10 23:57:28 | INFO     | transformation | item_genre_matrix     : 3883 rows x 20 cols  (19 genres)
2026-11-10 23:57:28 | INFO     | transformation | user_genre_profile    : 5392 rows x 20 cols
2026-11-10 23:57:28 | INFO     | transformation | user_item_matrix      : 5392 x 3662 sparse CSR, nnz=799,725, density=4.05%
2026-11-10 23:57:28 | INFO     | transformation | Sample user_activity_summary:
 user_id  n_interactions  mean_rating  std_rating  pct_high_ratings  first_ts   last_ts  active_days  tenure_days  days_since_last
     635              36         4.36        0.59              0.94 975767861 975768756            1         0.01             0.00
     636              86         4.06        0.91              0.71 975751854 975753027            1         0.01             0.18
     637              37         3.76        1.08              0.65 975733398 975739614            1         0.07             0.34
2026-11-10 23:57:28 | INFO     | transformation | Top-5 item_popularity:
 item_id                                clean_title  n_ratings  mean_rating  bayes_rating  popularity_rank
    2858                           American Beauty       2901        4.329         4.324                1
     260             Star Wars: Episode IV - A New Hope       2516        4.452         4.446                2
    1196 Star Wars: Episode V - The Empire Strikes Back       2516        4.290         4.285                2
    1210     Star Wars: Episode VI - Return of the Jedi       2453        4.019         4.016                4
     589                    Terminator 2: Judgment Day       2282        4.067         4.063                5
2026-11-10 23:57:28 | INFO     | transformation | Transformation finished in 0.68s
2026-11-10 23:57:28 | INFO     | orchestrator   | Run summary (0.7s total):
2026-11-10 23:57:28 | INFO     | orchestrator   |   transform  SUCCESS (0.7s)
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline %
```

A4 - Stage 4, Data transformation: derived datasets (activity summary, item popularity, genre matrices, sparse user-item matrix).

```
python run_pipeline.py --stage features
2026-11-11 00:02:43 | INFO     | orchestrator   | Execution plan: features
2026-11-11 00:02:43 | INFO     | features       | ============================================================================
2026-11-11 00:02:43 | INFO     | features       | STAGE: 5. FEATURE MANAGEMENT  (registry → feature store → catalog)
2026-11-11 00:02:43 | INFO     | features       | ============================================================================
2026-11-11 00:02:43 | INFO     | features       | Registry check passed: all 28 declared features materialised
2026-11-11 00:02:43 | INFO     | features       | user_features : 5,392 users x 32 columns
2026-11-11 00:02:43 | INFO     | features       | item_features : 3,883 items x 29 columns
2026-11-11 00:02:43 | INFO     | features       | Feature catalog (28 features) -> reports/feature_catalog.md
2026-11-11 00:02:43 | INFO     | features       | FeatureStore.get_user_features([635, 639]):
         user_activity_count  user_avg_rating user_preferred_genre  user_days_since_last_interaction
user_id
635                       36           4.361              Drama                             0.000
639                       79           3.696             Action                             0.459
2026-11-11 00:02:43 | INFO     | features       | FeatureStore.get_item_features([2858, 260]):
         item_popularity  item_avg_rating  item_bayes_rating  item_popularity_rank
item_id
2858                2901            4.329              4.324                     1
260                 2516            4.452              4.446                     2
2026-11-11 00:02:43 | INFO     | features       | FeatureStore.get_user_item_features(...):
 user_id  item_id  ui_genre_affinity  ui_rating_gap  ui_popularity_vs_user
     635     2858              0.873         -0.037                  0.204
     635      260              0.083          0.084                  0.203
2026-11-11 00:02:43 | INFO     | features       | Feature build finished in 0.11s
2026-11-11 00:02:43 | INFO     | orchestrator   | Run summary (0.1s total):
2026-11-11 00:02:43 | INFO     | orchestrator   |   features   SUCCESS (0.1s)
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline %
```

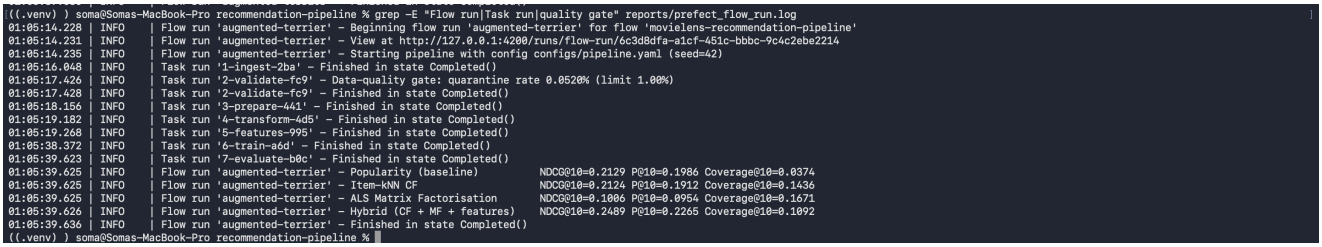
A5 - Stage 5, Feature management: registry check, feature store tables, catalog, FeatureStore API calls.

```
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline % python run_pipeline.py --stage train
2026-11-11 00:06:42 | INFO     | orchestrator   | Execution plan: train
2026-11-11 00:06:42 | INFO     | training       | ============================================================================
2026-11-11 00:06:42 | INFO     | training       | STAGE: 6. MODEL TRAINING  (tuning on validation split + MLflow tracking)
2026-11-11 00:06:42 | INFO     | training       | ============================================================================
2026/11/11 00:06:45 INFO mlflow.store.db.utils: Creating initial MLflow database tables...
2026/11/11 00:06:45 INFO mlflow.store.db.utils: Updating database tables
2026-11-11 00:06:45 | INFO     | training       | Tuning on validation split: sub-train=719,751 rows, validation=22,725 rows, 360 users
2026-11-11 00:06:46 | INFO     | training       |   item_knn  neighbours=50   shrinkage=20   val_ndcg_at_10=0.1149
2026-11-11 00:06:46 | INFO     | training       |   item_knn  neighbours=50   shrinkage=100  val_ndcg_at_10=0.1175
2026-11-11 00:06:47 | INFO     | training       |   item_knn  neighbours=100  shrinkage=20   val_ndcg_at_10=0.1172
2026-11-11 00:06:47 | INFO     | training       |   item_knn  neighbours=100  shrinkage=100  val_ndcg_at_10=0.1215
2026-11-11 00:06:48 | INFO     | training       |   item_knn  neighbours=300  shrinkage=20   val_ndcg_at_10=0.1344
2026-11-11 00:06:48 | INFO     | training       |   item_knn  neighbours=300  shrinkage=100  val_ndcg_at_10=0.1386
2026-11-11 00:06:51 | INFO     | training       |   als_mf    reg=0.05               val_rmse=0.9035
2026-11-11 00:06:53 | INFO     | training       |   als_mf    reg=0.08               val_rmse=0.8844
2026-11-11 00:06:55 | INFO     | training       |   als_mf    reg=0.12               val_rmse=0.8878
2026-11-11 00:06:55 | INFO     | training       |   hybrid    {'item_knn': 0.55, 'als_mf': 0.2, 'genre_affinity': 0.15, 'popularity': 0.1} val_ndcg_at_10=0.1834
2026-11-11 00:06:55 | INFO     | training       |   hybrid    {'item_knn': 0.5, 'als_mf': 0.1, 'genre_affinity': 0.1, 'popularity': 0.3} val_ndcg_at_10=0.1743
2026-11-11 00:06:55 | INFO     | training       |   hybrid    {'item_knn': 0.4, 'als_mf': 0.1, 'genre_affinity': 0.1, 'popularity': 0.4} val_ndcg_at_10=0.1810
2026-11-11 00:06:55 | INFO     | training       |   hybrid    {'item_knn': 0.45, 'als_mf': 0.25, 'genre_affinity': 0.05, 'popularity': 0.25} val_ndcg_at_10=0.1883
2026-11-11 00:06:55 | INFO     | training       | Selected: item_knn={'neighbours': 300, 'shrinkage': 100} | als_mf reg=0.08 | hybrid weights={'item_knn': 0.45, 'als_mf': 0.25, 'genre_affinity': 0.05, 'popularity':
 0.25}
2026-11-11 00:06:55 | INFO     | training       | Hyper-parameter tuning finished in 10.37s
2026-11-11 00:06:55 | INFO     | training       | Trained popularity in   0.00s  params={'min_ratings': 20}  run_id=d95a2a5b31d8
2026-11-11 00:06:56 | INFO     | training       | Trained item_knn   in   0.48s  params={'neighbours': 300, 'shrinkage': 100}  run_id=4a2d0a414325
2026-11-11 00:06:58 | INFO     | training       | Trained als_mf     in   2.34s  params={'factors': 48, 'reg': 0.08, 'bias_reg': 10.0, 'iterations': 10}  run_id=c84a11b7ffdb final_train_rmse=0.7375
2026-11-11 00:06:58 | INFO     | training       | Trained hybrid     in   0.00s  params={'weights': {'item_knn': 0.45, 'als_mf': 0.25, 'genre_affinity': 0.05, 'popularity': 0.25}}  run_id=422f18cac64d
2026-11-11 00:06:58 | INFO     | training       | Final model training finished in 2.87s
2026-11-11 00:06:58 | INFO     | training       | Model registry -> models/model_registry.json ; MLflow experiment 'movielens-recsys'
2026-11-11 00:06:58 | INFO     | orchestrator   | Run summary (16.7s total):
2026-11-11 00:06:58 | INFO     | orchestrator   |   train      SUCCESS (16.7s)
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline %
```

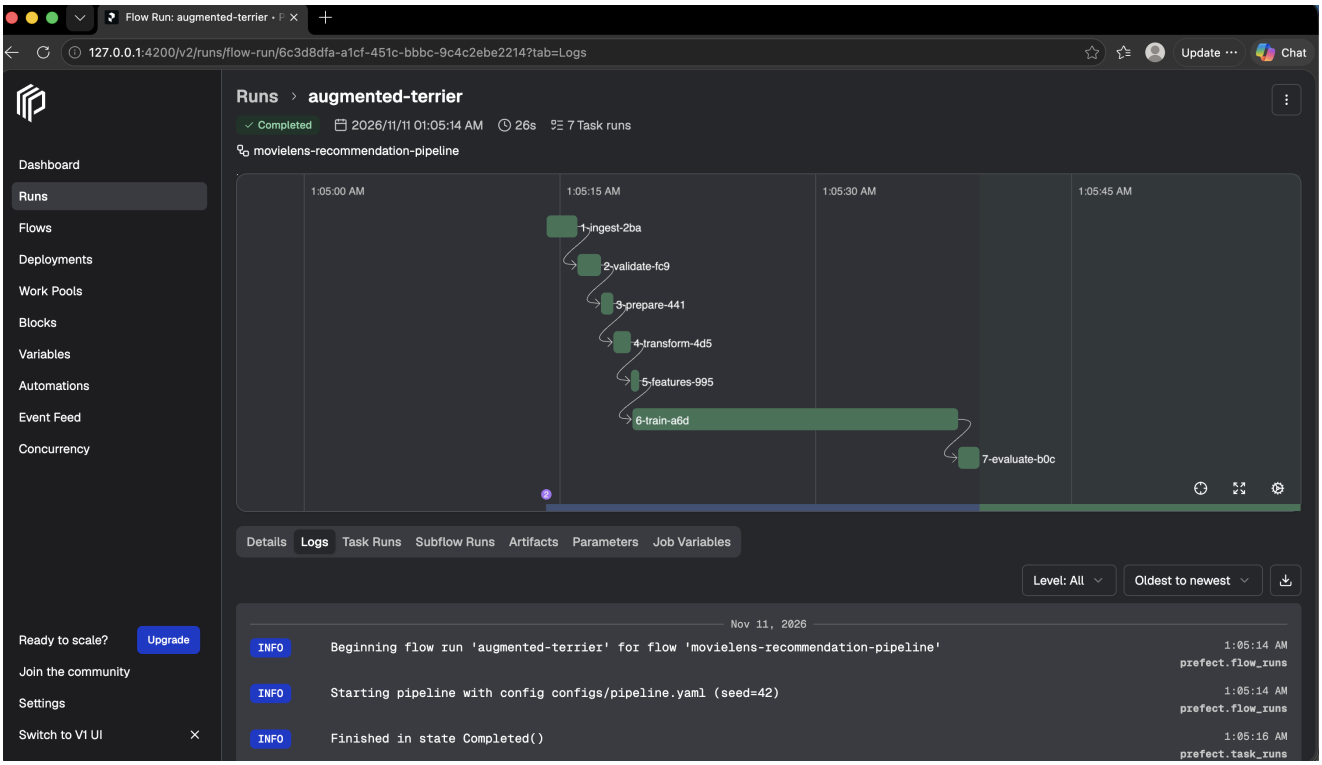
A6 - Stage 6, Model training: validation-split tuning (MLflow nested runs) and final training of the four models.

```
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline % python run_pipeline.py --stage evaluate
2026-11-11 00:41:42 | INFO     | orchestrator   | Execution plan: evaluate
2026-11-11 00:41:42 | INFO     | evaluation     | ============================================================================
2026-11-11 00:41:42 | INFO     | evaluation     | STAGE: 7. MODEL EVALUATION  (P/R/NDCG@K, coverage, diversity, RMSE/MAE)
2026-11-11 00:41:42 | INFO     | evaluation     | ============================================================================
2026-11-11 00:41:43 | INFO     | evaluation     | Test period: 103,783 ratings | 1135 users (1114 with >=1 relevant item, rating >= 4) | catalogue=3662 items
2026-11-11 00:41:43 | INFO     | evaluation     | Popularity (baseline)          P@10=0.1986 R@10=0.0479 NDCG@10=0.2129 Cov=0.0374 Div=0.7601 RMSE=0.9595 MAE=0.7597
2026-11-11 00:41:43 | INFO     | evaluation     | Item-kNN CF                    P@10=0.1912 R@10=0.0548 NDCG@10=0.2124 Cov=0.1436 Div=0.7370 RMSE=- MAE=-
2026-11-11 00:41:43 | INFO     | evaluation     | ALS Matrix Factorisation       P@10=0.0954 R@10=0.0256 NDCG@10=0.1006 Cov=0.1671 Div=0.7904 RMSE=0.8941 MAE=0.6956
2026-11-11 00:41:44 | INFO     | evaluation     | Hybrid (CF + MF + features)    P@10=0.2265 R@10=0.0646 NDCG@10=0.2489 Cov=0.1092 Div=0.7443 RMSE=- MAE=-
2026-11-11 00:41:44 | INFO     | evaluation     | Evaluation finished in 0.93s
2026-11-11 00:41:45 | INFO     | evaluation     | Comparison table (K=10):
                       Precision@10  Recall@10  NDCG@10  HitRate@10  Coverage@10  Diversity@10  Novelty@10    RMSE     MAE
Model
Popularity (baseline)            0.1986     0.0479   0.2129      0.6284       0.0374        0.7601      1.4128  0.9595  0.7597
Item-kNN CF                      0.1912     0.0548   0.2124      0.6221       0.1436        0.7370      1.9984     NaN     NaN
ALS Matrix Factorisation         0.0954     0.0256   0.1006      0.4434       0.1671        0.7904      4.3955  0.8941  0.6956
Hybrid (CF + MF + features)      0.2265     0.0646   0.2489      0.6732       0.1092        0.7443      1.8716     NaN     NaN
2026-11-11 00:41:45 | INFO     | evaluation     | Best model by NDCG@10: Hybrid (CF + MF + features)
2026-11-11 00:41:45 | INFO     | evaluation     | Reports -> reports/model_comparison.{md,csv,json}, reports/figures/model_comparison.png
2026-11-11 00:41:45 | INFO     | orchestrator   | Run summary (2.4s total):
2026-11-11 00:41:45 | INFO     | orchestrator   |   evaluate   SUCCESS (2.4s)
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline %
```

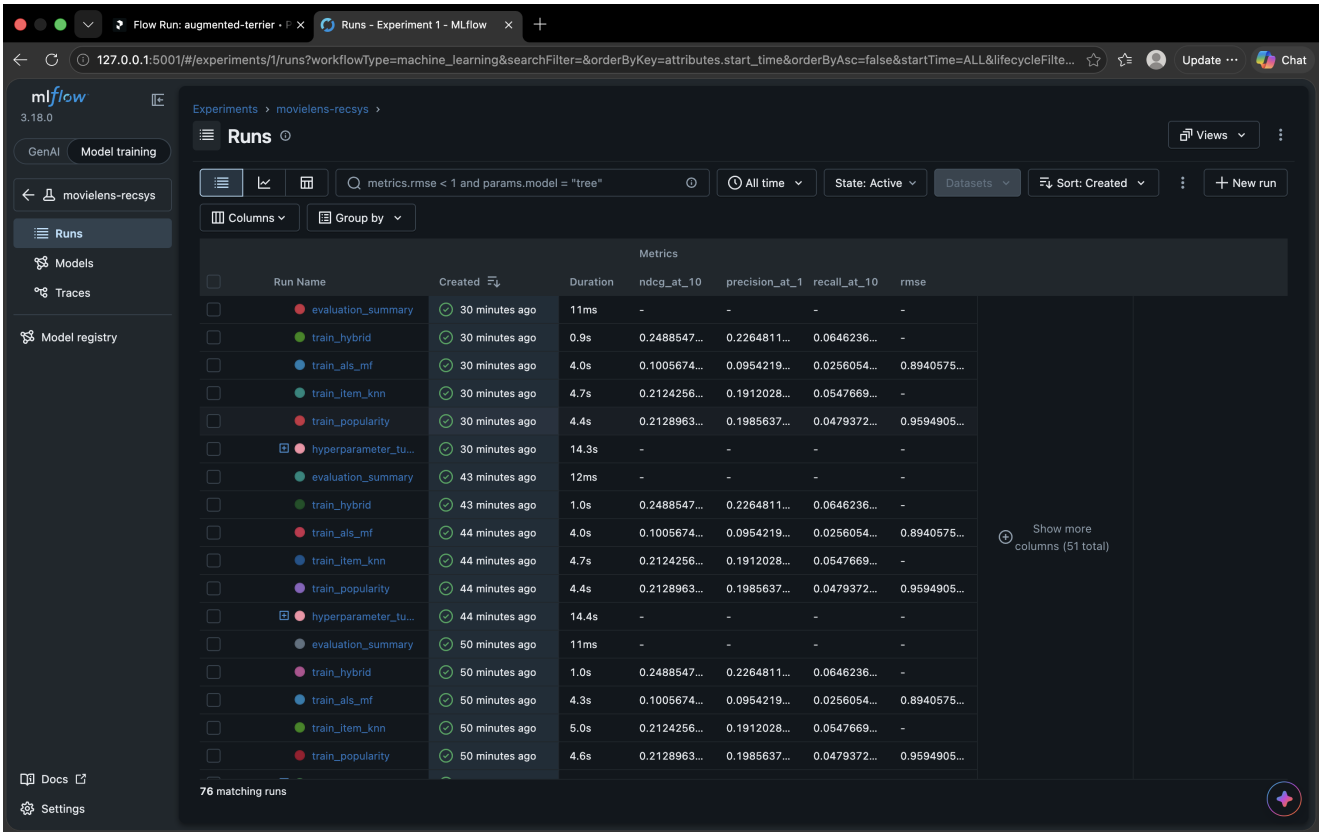
A7 - Stage 7, Model evaluation: P/R/NDCG@10, coverage, diversity, novelty, RMSE/MAE comparison.



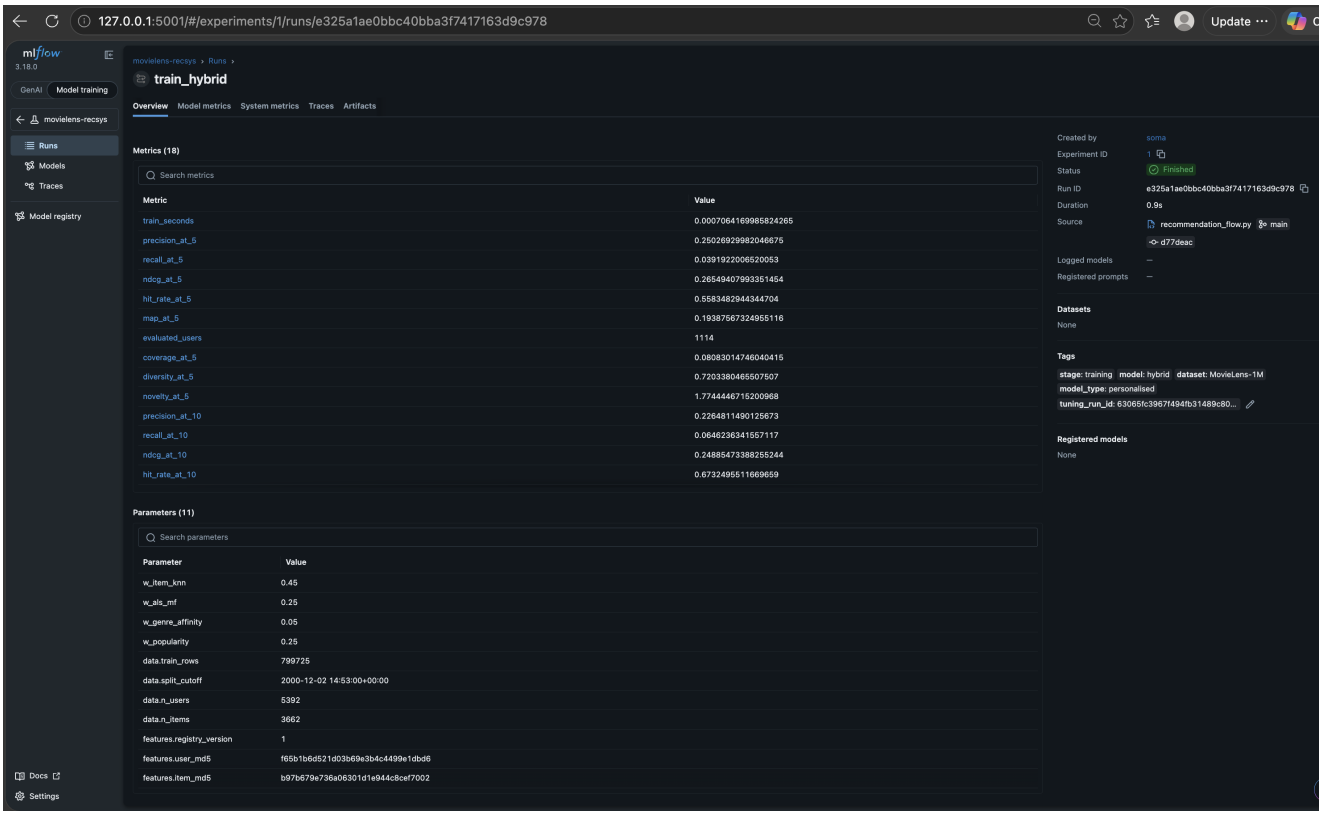
A8 - Orchestration: Prefect flow run summary - the seven tasks completed in dependency order, data-quality gate and final model scores.



A9 - Prefect UI: successful flow run, task dependency graph and logs (7/7 tasks Completed).



A10 - MLflow experiment 'movielens-recsys': tuning parent runs, training runs and evaluation summaries, one set per pipeline execution.



A11 - MLflow run detail (train_hybrid): logged metrics, parameters, lineage tags and link to the tuning run.

```
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline % python scripts/show_mlflow_runs.py
MLflow tracking URI: sqlite:////Users/soma/Downloads/recommendation-pipeline/mlflow.db
Experiment: movielens-recsys  |  latest pipeline execution (4 training runs)

Parameters:
      run_id      model neighbours shrinkage factors  reg min_ratings train_rows feat_ver
e325a1ae0bbc     hybrid          –         –       –    –           –     799725        1
d03de936e77f     als_mf          –         –      48 0.08          –     799725        1
fc4aef2c2c89   item_knn        300       100       –    –           –     799725        1
45884e0763ae popularity          –         –       –    –          20     799725        1

Metrics:
     model   P@10   R@10  NDCG@10  Cov@10  Div@10   RMSE    MAE  train_s
    hybrid 0.2265 0.0646   0.2489  0.1092  0.7443      –      –   0.0007
    als_mf 0.0954 0.0256   0.1006  0.1671  0.7904 0.8941 0.6956   3.4150
  item_knn 0.1912 0.0548   0.2124  0.1436  0.7370      –      –   0.7509
popularity 0.1986 0.0479   0.2129  0.0374  0.7601 0.9595 0.7597   0.0008

Artifacts of run e325a1ae0bbc: ['lineage', 'model']
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline %
```

A12 - Tracked parameters and metrics of the latest execution, queried from MLflow.

```
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline % python -m src.serving.recommend --user 635 --model hybrid --k 10
User profile (from feature store):
         user_activity_count  user_avg_rating user_preferred_genre  user_mainstreaminess
user_id
635                       36            4.361               Drama                 0.796

Top-10 recommendations for user 635 (model: hybrid):
 rank  item_id                                clean_title  release_year           genres  item_popularity  item_bayes_rating  ui_genre_affinity
    1      923                    Citizen Kane          1941            Drama              888              4.405              0.901
    2      912                      Casablanca          1942 Drama|Romance|War             1385              4.412              0.693
    3     1193 One Flew Over the Cuckoo's Nest          1975            Drama             1427              4.389              0.901
    4      111                      Taxi Driver          1976    Drama|Thriller             1038              4.204              0.660
    5     1225                          Amadeus          1984            Drama             1114              4.218              0.901
    6      318          The Shawshank Redemption          1994            Drama             1845              4.549              0.901
    7     1247                   The Graduate           1967     Drama|Romance             1004              4.226              0.755
    8      527                Schindler's List          1993         Drama|War             1933              4.493              0.731
    9     1084                Bonnie and Clyde          1967       Crime|Drama              568              4.078              0.707
   10      908               North by Northwest         1959    Drama|Thriller             1100              4.386              0.660
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline %
```

A13 - Prediction: top-10 recommendations for user 635 from the hybrid model, with features read from the same feature store.

```
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline % python -m pytest -q tests
.....
5 passed in 1.06s
((.venv) ) soma@Somas-MacBook-Pro recommendation-pipeline %
```

A14 - Unit tests for metrics and DAG ordering.